

Gauge-Free Weighted Riesz Terms for Nyström Peripheral Extraction

A Second-Order Continuum Bridge and Exact Stored Haar Ledgers

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Abstract

A nested-grid spectral-count program subtracts a rank-two Perron/parity term from folded-Gaussian Markov matrices. Smooth-kernel Haar theory predicts second-order consistency and detail blocks and first-order cross blocks, but the missing projector bridge was previously phrased in terms of individual left and right eigenvectors. Those vectors have unavoidable scaling, sign, phase, and ordering ambiguities. We replace them by the intrinsic weighted Riesz term

$$\mathcal{Q}(A; \Gamma) = A\Pi(A; \Gamma) = \frac{1}{2\pi i} \int_{\Gamma} z(z - A)^{-1} dz.$$

First, for two bounded operators sharing an isolating contour, we prove the exact perturbation estimate

$$\|\mathcal{Q}(A; \Gamma) - \mathcal{Q}(B; \Gamma)\| \leq \frac{\text{len}(\Gamma)}{2\pi} \max_{z \in \Gamma} |z| M_A M_B \|A - B\|,$$

where M_A and M_B are the contour resolvent suprema. Second, for a C^2 integral kernel and a simple isolated nonzero eigenvalue, midpoint Nyström approximation of the eigenvalue and of the right and transposed left eigenfunctions yields an intrinsic matrix estimate

$$\|Q_h - Q_h^\circ\|_2 = O(h^2).$$

Here Q_h is the finite-dimensional weighted Riesz term and Q_h° is the midpoint matrix of the smooth continuum kernel $\lambda r(x)\ell(y)$. The argument is independent of every eigenvector gauge. Strong positivity makes the Perron instance unconditional for the full folded-Gaussian kernel. The negative parity instance remains conditional on the existence of a simple isolated continuum resonance. The adaptive cutoff bridge from the preceding paper transfers to Q_h under one additional, explicit uniform Euclidean contour-resolvent bound.

At $\sigma = 10^{-2}$ we then audit the exact stored binary64 rank-two factors at dimensions 2048, 4096, 8192. A 224-bit Arb calculation evaluates the low-rank Frobenius identities without materializing the dense matrices. The second-to-first Haar ratios are rigorously enclosed by

$$0.249816736, \quad 0.500028226, \quad 0.500040220, \quad 0.250029234$$

for the consistency, coarse-to-detail, detail-to-coarse, and detail blocks. The exact-stored biorthogonality defect is below 2.885×10^{-15} . Floating eigen-residuals and an observed radial gap of 0.31188 support, but do not validate, the parity-isolation hypothesis. Thus the Perron bridge is closed analytically; continuum parity isolation is the remaining analytic gate. No zero-noise, zeta-zero, Hilbert-Pólya, or Riemann-hypothesis conclusion is made.

1 Introduction

The stored continuation certificates of Wang [12, 13] use a folded-Gaussian Markov matrix P_h , remove its Perron and negative parity contributions, and square the remaining one-step operator. A subsequent deterministic theorem [11] explained the observed dyadic law

$$\|E_h\|_2, \|D_h\|_2 = O(h^2), \quad \|C_h\|_2, \|B_h\|_2 = O(h). \quad (1)$$

One Haar difference extracts one derivative, while two differences extract a mixed second derivative. Discrete row normalization, subtraction of a smooth finite-rank kernel, and exact squaring preserve these powers.

That result isolated two bridges between the smooth continuum theorem and the sparse stored family. The hard support window has now been treated separately [10]: a fixed eight-sigma window is negligible at the archived dimensions but has a nonzero full-kernel continuum defect, whereas a slowly growing support gives an $O(h^2)$ Euclidean perturbation. The remaining bridge concerns the peripheral term.

A direct comparison of eigenvectors is the wrong invariant. If r and ℓ are a biorthogonal right-left pair, then

$$r \mapsto cr, \quad \ell \mapsto c^{-1}\ell$$

changes both vectors but leaves the spectral contribution unchanged. Signs and complex phases are special cases, and several modes introduce ordering and basis ambiguities. Numerical continuation can impose conventions, but an analytic theorem should not depend on them.

The matrix actually subtracted in the stored construction is

$$Q_h = R_h \Lambda_h L_h^T, \quad L_h^T R_h = I, \quad (2)$$

not the unweighted projector $R_h L_h^T$. Intrinsically, it is the part of P_h selected by holomorphic functional calculus [6]:

$$\mathcal{Q}(P_h; \Gamma) = P_h \Pi(P_h; \Gamma). \quad (3)$$

This observation turns the missing bridge into a resolvent and Nyström question.

The contributions of this paper are:

1. a gauge-free definition of the weighted peripheral term for a simple eigenvalue or a finite cluster;
2. an explicit resolvent-Lipschitz estimate and a Neumann-series contour persistence criterion;
3. a second-order Euclidean matrix theorem for smooth midpoint Nyström weighted terms, proved by treating the transposed kernel on the same footing as the original kernel;
4. an unconditional application to the full-kernel Perron branch and a sharply stated conditional application to the negative parity branch;
5. transfer of an $O(h^2)$ Markov cutoff defect to the weighted Riesz term under uniform contour conditioning; and
6. an exact-stored Arb ledger for the rank-two factors and all four dyadic Haar blocks at the three archived dimensions.

Three evidence levels are kept distinct.

Analytic theorem

Functional-calculus perturbation bounds and smooth Nyström convergence.

Exact-stored ledger

Arb enclosures after every archived binary64 factor is treated as an exact real input.

Floating diagnostic

Sparse eigen-residuals and a leading-spectrum isolation audit; these are not interval enclosures of the eigensolver or of the full matrix.

The result closes an analytic mechanism, not an arithmetic identification. The resonances studied here are eigenvalues of a non-self-adjoint noisy transfer operator at fixed positive noise.

2 Weighted Riesz terms

Let A be a bounded operator on a complex Banach space, and let Γ be a positively oriented rectifiable Jordan contour contained in the resolvent set of A . Assume that its interior contains a finite isolated spectral cluster.

Definition 2.1 (Riesz projector and weighted term). Define

$$\Pi(A; \Gamma) = \frac{1}{2\pi i} \int_{\Gamma} (z - A)^{-1} dz, \quad (4)$$

$$\mathcal{Q}(A; \Gamma) = \frac{1}{2\pi i} \int_{\Gamma} z(z - A)^{-1} dz. \quad (5)$$

Proposition 2.2 (Intrinsic spectral contribution). *The weighted term satisfies*

$$\mathcal{Q}(A; \Gamma) = A\Pi(A; \Gamma) = \Pi(A; \Gamma)A.$$

If the contour encloses one simple eigenvalue λ , with $Ar = \lambda r$, $\ell A = \lambda \ell$, and $\ell(r) = 1$, then

$$\mathcal{Q}(A; \Gamma) = \lambda r \otimes \ell. \quad (6)$$

This expression is invariant under $r \mapsto cr$ and $\ell \mapsto c^{-1}\ell$ for every $c \neq 0$.

Proof. The identity $z(z - A)^{-1} = I + A(z - A)^{-1}$ and the vanishing contour integral of the constant operator give the first formula. The standard simple Riesz projector is $r \otimes \ell$, which proves the second. The two gauge factors cancel exactly. $\square$

For a semisimple cluster, one may write $\mathcal{Q} = R\Lambda L^T$ in any biorthogonal basis. A change of basis alters the factors but not $\mathcal{Q}$. The contour formula also remains valid for nonsemisimple clusters, where an eigenvector-only description would need Jordan data.

Theorem 2.3 (Resolvent-Lipschitz bound). *Let A and B be bounded operators on the same Banach space, and suppose $\Gamma \subset \rho(A) \cap \rho(B)$. Put*

$$M_A = \max_{z \in \Gamma} \|(z - A)^{-1}\|, \quad M_B = \max_{z \in \Gamma} \|(z - B)^{-1}\|,$$

and $R_{\Gamma} = \max_{z \in \Gamma} |z|$. Then

$$\|\mathcal{Q}(A; \Gamma) - \mathcal{Q}(B; \Gamma)\| \leq \frac{\text{len}(\Gamma)}{2\pi} R_{\Gamma} M_A M_B \|A - B\|. \quad (7)$$

Proof. The first resolvent identity gives

$$(z - A)^{-1} - (z - B)^{-1} = (z - A)^{-1}(A - B)(z - B)^{-1}.$$

Insert this into (5), take norms along the contour, and bound the Bochner integral by its arclength integral. $\square$

Corollary 2.4 (Contour persistence from one resolvent bound). *Let $\delta = \|A - B\|$ and assume $M_A \delta < 1$. Then $\Gamma \subset \rho(B)$,*

$$M_B \leq \frac{M_A}{1 - M_A \delta}, \quad (8)$$

and

$$\|\mathcal{Q}(A; \Gamma) - \mathcal{Q}(B; \Gamma)\| \leq \frac{\text{len}(\Gamma)}{2\pi} R_{\Gamma} \frac{M_A^2 \delta}{1 - M_A \delta}. \quad (9)$$

Proof. Factor

$$z - B = \{I - (B - A)(z - A)^{-1}\}(z - A).$$

The first factor is invertible by a uniformly convergent Neumann series. This proves (8); then apply [theorem 2.3](#). $\square$

The factor $M_A M_B$ is essential in a nonnormal problem. A geometric distance from Γ to the spectrum does not, by itself, replace it. This is exactly where a visually stable eigenvalue gap stops being a validated projector theorem.

3 A second-order Nyström bridge

Let $I = [0, 1]$ and $k \in C^2(I \times I)$. Define

$$(\mathcal{K}f)(x) = \int_0^1 k(x, y) f(y) dy. \quad (10)$$

For $h = 1/n$, use midpoint nodes $x_i = (i + \frac{1}{2})h$, $0 \leq i < n$, and the matrix

$$(K_h)_{ij} = hk(x_i, x_j). \quad (11)$$

Here T denotes algebraic transpose: l_h^T is the row eigenfunctional. For the archived real branches this agrees with the usual real left-vector notation; no Hermitian gauge convention is being imposed.

Suppose $\lambda \neq 0$ is a simple isolated eigenvalue of $\mathcal{K}$. Its right eigenfunction and left eigenfunctional have C^2 representatives r and ℓ because

$$r = \lambda^{-1} \mathcal{K}r, \quad \ell = \lambda^{-1} \mathcal{K}^T \ell,$$

where $(\mathcal{K}^T g)(y) = \int_0^1 k(x, y) g(x) dx$. Choose the bilinear normalization

$$\int_0^1 \ell(x) r(x) dx = 1. \quad (12)$$

The continuum weighted Riesz kernel is

$$q(x, y) = \lambda r(x) \ell(y), \quad (13)$$

and its midpoint matrix is

$$(Q_h^\circ)_{ij} = hq(x_i, x_j). \quad (14)$$

Theorem 3.1 (Gauge-free second-order simple branch). *For all sufficiently small h , K_h has a simple eigenvalue λ_h inside a fixed isolating contour for λ . Let $r_h, l_h \in \mathbb{C}^n$ be any corresponding right and left vectors normalized by $l_h^T r_h = 1$, and put*

$$Q_h = \lambda_h r_h l_h^T = \mathcal{Q}(K_h; \Gamma). \quad (15)$$

Then

$$\|Q_h - Q_h^\circ\|_2 = O(h^2). \quad (16)$$

The estimate is unchanged by every admissible rescaling of r_h, l_h .

Proof. Collectively compact Nyström theory gives spectral exactness for the nonzero isolated eigenvalue. Composite midpoint quadrature is second order on the smooth eigenfunctions.

Applied to k and to the transposed kernel $k^T(x, y) = k(y, x)$, the standard simple-branch estimates [1–4] give compatible node vectors ρ_h, η_h such that

$$|\lambda_h - \lambda| = O(h^2), \quad (17)$$

$$\max_i |\rho_{h,i} - r(x_i)| + \max_i |\eta_{h,i} - \ell(x_i)| = O(h^2), \quad (18)$$

with $K_h \rho_h = \lambda_h \rho_h$ and $K_h^T \eta_h = \lambda_h \eta_h$.

Set

$$a_h = h \sum_{i=0}^{n-1} \eta_{h,i} \rho_{h,i}.$$

By (12), (18), and the midpoint rule, $a_h = 1 + O(h^2)$. Hence we may choose

$$r_h = \rho_h, \quad l_h = \frac{h}{a_h} \eta_h,$$

which gives $l_h^T r_h = 1$. Uniform boundedness of r, ℓ and their approximants now implies, entrywise,

$$|\lambda_h(r_h)_i(l_h)_j - h\lambda r(x_i)\ell(x_j)| = O(h^3).$$

For an $n \times n$ matrix with maximum entry $O(h^3)$, both the 1- and ∞ -norms are $O(nh^3) = O(h^2)$; their geometric mean bounds the spectral norm. This proves (16). Finally, Q_h itself is invariant under the right-left gauge by proposition 2.2. $\square$

Corollary 3.2 (Finite simple cluster). *If a fixed finite cluster consists of simple isolated nonzero eigenvalues with disjoint isolating contours, the sum of their weighted terms satisfies the same $O(h^2)$ matrix estimate.*

Proof. Apply theorem 3.1 to each branch and add finitely many estimates. $\square$

Remark 3.3 (Why the result is stronger than vector tracking). The proof chooses one convenient normalization only to obtain an entrywise bound. The conclusion concerns Q_h , which is independent of that choice. Thus no sign-alignment, phase-alignment, or mode-ordering algorithm appears in the theorem statement.

3.1 Discretely normalized smooth kernels

The archived full Markov matrix uses a discrete row normalizer. Let a positive C^2 raw kernel g have continuum normalizer

$$Z(x) = \int_0^1 g(x, y) dy > 0$$

and normalized kernel $k(x, y) = g(x, y)/Z(x)$. Its full midpoint Markov matrix is

$$(P_h)_{ij} = \frac{g(x_i, x_j)}{\sum_m g(x_i, x_m)}. \quad (19)$$

Equivalently, P_h is the Nyström matrix of the h -dependent smooth kernel $g(x, y)/Z_h(x)$, where $Z_h(x) = h \sum_m g(x, x_m)$. Midpoint quadrature gives $Z_h - Z = O(h^2)$ uniformly.

Proposition 3.4 (Discrete row normalization). *If $\lambda \neq 0$ is a simple isolated eigenvalue of the continuum-normalized operator with kernel $k = g/Z$, then its weighted term for the matrices P_h in (19) is $O(h^2)$ close in spectral norm to the midpoint matrix of the continuum weighted kernel.*

Proof. Positivity and compactness separate Z uniformly from zero. Composite midpoint quadrature and $g \in C^2$ give

$$\|Z_h - Z\|_\infty = O(h^2), \quad \|g/Z_h - g/Z\|_\infty = O(h^2).$$

The h -dependent kernels g/Z_h and their transposes remain uniformly smooth and collectively compact. The simple right and transposed-left Nyström branches therefore satisfy the estimates (17)–(18); repeating the normalization and entrywise argument in [theorem 3.1](#) proves the claim. $\square$

Corollary 3.5 (Full folded-Gaussian Perron bridge). *Fix $\sigma > 0$ and a smooth map f . On $I = [0, 1]$, let*

$$g_\sigma(x, y) = \exp\left[-\frac{(y - f(x))^2}{2\sigma^2}\right] + \exp\left[-\frac{(y + f(x))^2}{2\sigma^2}\right]. \quad (20)$$

The continuum-normalized operator is compact, Markov, and strongly positive. Its eigenvalue 1 is algebraically simple and isolated. Consequently the weighted Perron term of the full discretely normalized midpoint matrices satisfies (16) without an additional spectral hypothesis.

Proof. The normalized kernel is strictly positive and smooth on the compact square. Strong positivity and Krein–Rutman theory make 1 algebraically simple and exclude every other peripheral spectral value [8, 9]. Apply [proposition 3.4](#). $\square$

Corollary 3.6 (Conditional parity bridge). *If the full folded-Gaussian continuum operator has a simple isolated nonzero negative resonance λ_- continuing the archived parity branch, then its weighted Nyström term is $O(h^2)$ close in spectral norm to the midpoint matrix of $\lambda_- r_-(x)\ell_-(y)$. The combined Perron/parity rank-two term has the same rate.*

The hypothesis in [corollary 3.6](#) is not proved by three convergent-looking matrix eigenvalues. Establishing it requires either a validated continuum contour resolvent or an equivalent analytic isolation argument.

4 Transfer through a Gaussian support cutoff

Let P_h denote the full discretely normalized folded-Gaussian matrix and $\hat{P}_h$ a support-truncated and renormalized version of the same dimension. Suppose

$$\|\hat{P}_h - P_h\|_2 \leq \varepsilon_h. \quad (21)$$

For a fixed cluster contour Γ , put

$$M_h = \max_{z \in \Gamma} \|(z - P_h)^{-1}\|_2.$$

Proposition 4.1 (Markov-to-Riesz transfer). *If $\sup_h M_h \leq M < \infty$ and $M\varepsilon_h < 1$ for all sufficiently small h , then*

$$\|Q(\hat{P}_h; \Gamma) - Q(P_h; \Gamma)\|_2 \leq \frac{\text{len}(\Gamma)}{2\pi} R_\Gamma \frac{M^2 \varepsilon_h}{1 - M\varepsilon_h}. \quad (22)$$

In particular, an $O(h^2)$ Markov bridge gives an $O(h^2)$ weighted-Riesz bridge.

Proof. Apply [corollary 2.4](#) at each dimension. $\square$

The adaptive support schedule of Wang [10],

$$L(h) = \max\{5, 2\sqrt{\log(1/h)}\}, \quad (23)$$

gives

$$\varepsilon_h = O\left(h^2(\log(1/h))^{-1/4}\right).$$

Thus it transfers through [proposition 4.1](#) under the displayed uniform resolvent condition. A fixed eight-sigma window is extraordinarily small at the archived grids, but it does not converge to the full kernel in row-operator norm when held fixed forever. The finite-grid observation and the all-grid theorem are therefore different statements.

For the Perron branch, [corollary 3.5](#) closes the full smooth continuum bridge unconditionally. For the parity branch, two logically separate conditions remain: simple continuum isolation for [corollary 3.6](#), and uniform Euclidean contour conditioning for transfer through the nonsmooth cutoff. The latter may follow from a sufficiently strong validated version of the former, but it is not inferred from eigenvalue locations alone.

5 Exact-stored rank-two Haar ledger

We now study the binary64 factors archived in the two dyadic continuation papers. This section does not assert that they are exact eigenvectors of an exact-real sparse matrix. Instead, every stored binary64 number is treated as an exact real input and all subsequent low-rank identities are enclosed by Arb interval arithmetic [5, 7].

At dimension n , the stored term is

$$Q_n = R_n \Lambda_n L_n^T, \quad R_n, L_n \in \mathbb{R}^{n \times 2}. \quad (24)$$

For a fine level $2n$, define paired right averages and differences

$$\overline{R}_i = \frac{(R_{2n})_{2i} + (R_{2n})_{2i+1}}{2}, \quad \delta R_i = \frac{(R_{2n})_{2i} - (R_{2n})_{2i+1}}{2},$$

and paired left sums and differences

$$\Sigma L_i = (L_{2n})_{2i} + (L_{2n})_{2i+1}, \quad \Delta L_i = (L_{2n})_{2i} - (L_{2n})_{2i+1}.$$

The exact unnormalized Haar blocks are then

$$E_n = \overline{R} \Lambda_{2n} (\Sigma L)^T - R_n \Lambda_n L_n^T, \quad (25)$$

$$C_n = \delta R \Lambda_{2n} (\Sigma L)^T, \quad (26)$$

$$B_n = \overline{R} \Lambda_{2n} (\Delta L)^T, \quad (27)$$

$$D_n = \delta R \Lambda_{2n} (\Delta L)^T. \quad (28)$$

Thus E_n has rank at most four and the other blocks rank at most two. No dense $n \times n$ block is needed.

For real low-rank factors A, B ,

$$\|AB^T\|_F^2 = \text{tr}\left((A^T A)(B^T B)\right). \quad (29)$$

The implementation forms both small Gram matrices using exact binary64 inputs inside 224-bit Arb balls. This matters most for E_n , whose two rank-two contributions nearly cancel; a binary64 evaluation of the small Gram identity loses visible digits even though no large dense matrix is formed.

5.1 Rigorous block enclosures

Table 1 gives outward-rounded Frobenius intervals. Their widths are much smaller than the printed precision.

Table 1: 224-bit Arb enclosures for exact-stored weighted-term Haar block Frobenius norms.

block	2048 $\rightarrow$ 4096	4096 $\rightarrow$ 8192
E	[8.350827302101823, 8.350827302101826] 10^{-5}	[2.086176416513942, 2.086176416513943] 10^{-5}
C	[2.574357393980523, 2.574357393980524] 10^{-3}	[1.287251360405133, 1.287251360405134] 10^{-3}
B	[3.525041944225770, 3.525041944225772] 10^{-3}	[1.762662749289979, 1.762662749289980] 10^{-3}
D	[6.244505599125109, 6.244505599125111] 10^{-6}	[1.561308946412573, 1.561308946412575] 10^{-6}

Dividing the second interval by the first with outward rounding gives table 2. This is a rigorous property of the exact stored factors, not an asymptotic theorem inferred from two ratios.

Table 2: Outward-rounded exact-stored second-to-first Frobenius ratios.

block	ratio enclosure
E	[0.2498167356411347, 0.2498167356411349]
C	[0.5000282258458135, 0.5000282258458139]
B	[0.5000402199971905, 0.5000402199971907]
D	[0.2500292331599994, 0.2500292331599995]

Table 3: Outward-rounded renormalized exact-stored Frobenius constants. E, D are divided by h^2 and C, B by h .

block	2048 $\rightarrow$ 4096	4096 $\rightarrow$ 8192
E	[350.2590835651, 350.2590835652]	[350.0023235396, 350.0023235397]
C	[5.272283942872, 5.272283942873]	[5.272581572219, 5.272581572220]
B	[7.219285901774, 7.219285901775]	[7.219866621091, 7.219866621092]
D	[26.19135481243, 26.19135481244]	[26.19441743669, 26.19441743670]

The two second-order ratios lie within 1.84×10^{-4} and 2.93×10^{-5} of one quarter. The two first-order ratios lie within 2.83×10^{-5} and 4.03×10^{-5} of one half. This is the exact stored-factor counterpart of the smooth theorem in (1).

5.2 Biorthogonality and floating isolation audit

The Arb ledger also encloses $L_n^T R_n - I$ directly. Sparse eigen-residuals and the leading spectrum are then recomputed in binary64 only as diagnostics.

Table 4: Stored branch and isolation audit. Only the biorthogonality column is an exact-stored Arb upper; the eigenvalues, residuals, and observed bulk are floating diagnostics.

n	$\lambda_{-,n}$	$\ L_n^T R_n - I\ _2$ upper	max residual	parity/bulk radial gap
2048	-0.9865534036028739	$2.0070 \cdot 10^{-15}$	$3.307 \cdot 10^{-15}$	0.3118841
4096	-0.9865481927458079	$2.8849 \cdot 10^{-15}$	$3.516 \cdot 10^{-15}$	0.3118789
8192	-0.9865468907775788	$1.3674 \cdot 10^{-15}$	$3.395 \cdot 10^{-15}$	0.3118776

The successive parity increments are exactly evaluated from the stored binary64 eigenvalues inside Arb balls. Their ratio is enclosed by

$$0.24985683017206403 \leq \frac{\lambda_{8192} - \lambda_{4096}}{\lambda_{4096} - \lambda_{2048}} \leq 0.24985683017206410. \quad (30)$$

The two second-order Richardson extrapolates are

$$R_1 = -0.9865464557934524962 \dots, \quad (31)$$

$$R_2 = -0.9865464567881691756 \dots, \quad (32)$$

with

$$|R_2 - R_1| \leq 9.947166793959168 \times 10^{-10}. \quad (33)$$

The largest stored weighted-term spectral norm is 1.24055, and the largest unweighted projector spectral norm is 1.14810. These moderate values and the stable observed radial gap explain why ordinary floating continuation is well behaved on the three grids. They do not bound the resolvent on a continuum contour.

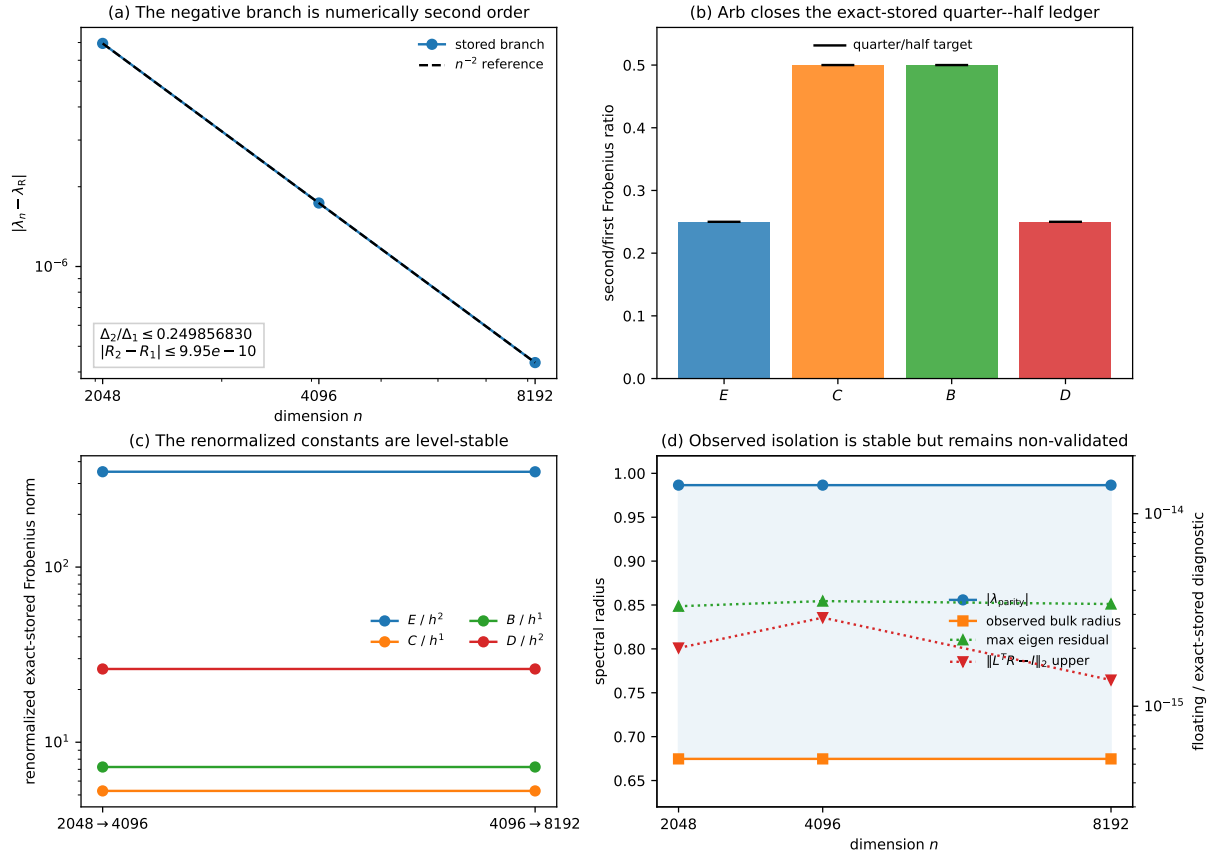


Figure 1: Weighted-Riesz bridge diagnostics. (a) The archived negative branch is consistent with second-order Nyström convergence, including two nearly coincident Richardson extrapolates. (b) Arb intervals close the exact-stored quarter-half Frobenius ledger. (c) Division by the predicted powers leaves nearly level-independent constants. (d) The negative branch remains separated from the twelve-mode observed bulk while eigen-residual and biorthogonality diagnostics stay near machine precision. Panel (d) is not a validated continuum isolation result.

6 What is closed and what remains

The projector bridge can now be stated without conflating evidence levels.

1. **Weighted Riesz stability is analytic.** [Theorem 2.3](#) is an exact operator theorem and includes the nonnormal conditioning explicitly.
2. **The full smooth Nyström bridge is analytic.** A simple isolated nonzero branch gives $\|Q_h - Q_h^\circ\|_2 = O(h^2)$ without choosing a persistent eigenvector gauge.
3. **The Perron instance is unconditional.** Strong positivity supplies simplicity and isolation for the full folded-Gaussian continuum operator.
4. **The parity instance is conditional.** The same conclusion follows if the archived negative branch is shown to continue a simple isolated continuum resonance. The current eigenvalue table does not prove that premise.
5. **The cutoff transfer is conditional on contour conditioning.** The adaptive $O(h^2)$ matrix bridge propagates to Q_h once a uniform Euclidean contour-resolvent bound is supplied.
6. **The stored Haar ledger is rigorous for its stated inputs.** The Arb intervals prove exact algebraic properties of the stored binary64 factors. Their status as spectral factors of the stored sparse matrices is still supported only by floating residuals.

Accordingly, the single remaining *analytic* gate in the rank-two continuum bridge is parity simplicity and isolation, preferably in a form that also controls the contour resolvent needed by [proposition 4.1](#). A fully validated finite-grid statement would add two computational gates: interval enclosure of the sparse eigensolver and outward-rounded construction of the binary64 Gaussian matrices.

Only two dyadic transitions are archived. They strongly identify the quarter-half mechanism but cannot prove a uniform all-level constant by themselves. The present paper also does not address the later hierarchical Schur-resolvent recursion, a zero-noise limit, a self-adjoint generator, or any arithmetic spectral identification.

7 Archive and reproducibility

The repository contains the exact-stored factors only through hashes of the RH-36 and RH-37 snapshots; the snapshots themselves remain in their source paper directories. The RH-40 certificate records those hashes together with the RH-38 Haar and RH-39 cutoff certificates.

The fast replay is

```
python -m pytest -q -p no:cacheprovider
python experiments/run_projector_pilot.py
python experiments/build_projector_certificate.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf weighted-riesz-projector-bridge.pdf
python experiments/build_archive.py
python experiments/verify_archive.py
```

The pilot recomputes sparse residuals and leading eigenvalues. The certificate then treats the resulting archived factors as exact inputs and uses the low-rank identity (29). The archive verifier checks dependency hashes, formulas, interval gates, publication artifacts, and the explicit limitations.

8 Conclusion

The appropriate continuum object for peripheral subtraction is not a chosen pair of eigenvectors but the weighted Riesz term AII. This replacement removes every gauge convention and exposes the only perturbative quantity that a nonnormal theorem genuinely needs: the contour resolvent.

For a smooth simple Nyström branch, the weighted term is second-order close in Euclidean matrix norm to the midpoint sample of its continuum kernel. Strong positivity closes this statement for the Perron branch of the full folded-Gaussian operator. The same theorem is ready for the archived negative branch as soon as its continuum simplicity and isolation are established. Under uniform contour conditioning, the adaptive Gaussian cutoff passes through the weighted Riesz calculus at the same order.

The exact-stored rank-two factors independently obey the predicted dyadic quarter-half law, with Arb-enclosed ratios within 2×10^{-4} of their ideal values and biorthogonality defects below 3×10^{-15} . The maze has therefore narrowed again: vector normalization is no longer an obstruction, and the next mathematically meaningful target is a validated continuum contour for the negative resonance.

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